Critical Analysis:
SPUQ: Perturbation-Based Uncertainty Quantification for Large Language Models

Kai-Yu Lu

1 Methodological Strengths

1.1 Problem framing and methodological scope

The study formulates uncertainty quantification (UQ) for large language models (LLMs) as a calibration problem under realistic deployment constraints, specifically the absence of reference answers at inference time and limited access to token-level probabilities in many closed APIs. The method explicitly targets both aleatoric uncertainty (inherent output variability) and epistemic uncertainty (knowledge or modeling gaps), addressing a concrete gap in prior LLM UQ work that is largely centered on aleatoric signals.

1.2 Core design choices and why they are effective

- **Input-side perturbation for epistemic sensitivity.** The perturbation module operates on temperature and prompts, a design that is feasible under closed LLM APIs where Monte Carlo Dropout style internal perturbations are unavailable. This choice directly operationalizes epistemic uncertainty as prediction instability under meaning-preserving perturbations, which is aligned with the motivating failure mode of confidently wrong predictions.
- **Sampling for aleatoric uncertainty with explicit aggregation.** Sampling multiple outputs remains a standard approach for capturing variability. The study strengthens this baseline by introducing an aggregation module that extends beyond exact match, thereby improving applicability to general text generation tasks.
- **Generalized agreement via textual similarity.** The inter-sample aggregation generalizes majority voting by replacing exact match with textual similarity metrics and introducing sample weights based on prompt similarity to attenuate the impact of extreme perturbations. This design directly addresses the known mismatch between exact match criteria and open-ended generation.
- **API-agnostic option via intra-sample aggregation.** The intra-sample pathway allows confidence to be computed either from likelihood-based signals when token probabilities exist or from verbalized uncertainty when they do not. This modularity broadens applicability across model families and access regimes.

1.3 Evaluation protocol and reporting

- **Multi-model, multi-dataset evaluation.** The experiments cover five LLMs spanning GPT and PaLM series and evaluate on one summarization dataset and four question answering datasets, including both classification-style and generation-style settings. This improves external validity relative to single-model reports.

- **Calibration-centric metrics.** The primary metric is Expected Calibration Error (ECE), complemented by Pearson correlation between confidence and accuracy, aligning evaluation with the core claim of improved uncertainty calibration.
- **Systematic study of design axes.** The study investigates multiple perturbation strategies (paraphrasing, dummy tokens, system message perturbation) and multiple similarity metrics (RougeL, BERTScore, SentenceBERT cosine similarity), enabling attribution of which choices consistently improve calibration.
- **Robustness check for hyperparameter tuning.** Hyperparameters are tuned on small development subsets with five independent tuning runs, and performance stability across development sets is explicitly reported, reducing concerns that improvements are driven by a single tuning split.

1.4 Ablation-style evidence and interpretability of effects

A case study demonstrates the central mechanism: a single confidently wrong answer under the original prompt is exposed by paraphrased prompts that induce output flips, producing a lower confidence score consistent with epistemic uncertainty. Additionally, comparisons against sampling without perturbation are interpreted as ablation-style evidence that the perturbation module drives the calibration gains.

1.5 Transparency and reproducibility measures

The paper provides a clear algorithmic specification of SPUQ, enumerates model endpoints and perturbation options, and reports key hyperparameter ranges for temperature perturbation. A public repository link is provided, supporting reproducibility.

2 Key Limitations

2.1 Evaluation scope and representativeness

- **Task diversity remains limited.** The evaluation emphasizes datasets where correctness is readily assessed via exact match or established F1 scoring. The paper explicitly notes that tasks with less well-defined accuracy, such as conversation and open-ended content generation, are not covered, limiting claims about real-world assistant behavior.
- **Summarization correctness definition is under-specified.** Calibration requires a binary or scalar notion of correctness. For summarization, the exact operationalization of accuracy used for ECE and correlation is not specified in the main text, creating ambiguity about what the confidence scores are calibrated to.

2.2 Perturbation validity and semantic fidelity

- **Meaning preservation is assumed, not verified.** Prompt perturbations are intended to be lexical variations that retain meaning, but the study does not report a systematic check that paraphrases preserve semantics across all datasets and queries. This can confound epistemic uncertainty with perturbation-induced task changes.

- **System message perturbation is narrow.** The implementation and analysis are acknowledged as constrained to basic system messages, leaving uncertainty about behavior under complex system instructions that are common in production settings.

2.3 Baselines and methodological completeness

- **Baseline coverage is incomplete for closed models.** Likelihood-based uncertainty is only applicable to GPT-3, limiting direct comparison between SPUQ and likelihood baselines on the strongest closed models that motivated the API-constraint argument.
- **Ablation is indirect.** The attribution to perturbation is largely inferred from comparisons against sampling without perturbation and descriptive statements. A fully controlled factorial ablation quantifying the marginal contribution of each perturbation type and each aggregation choice under fixed settings is not presented as a dedicated table.
- **Error analysis is limited.** The paper provides a motivating example and calibration plots, but it does not provide a detailed failure taxonomy for when perturbations degrade calibration, such as cases where paraphrasing changes question type, introduces ambiguity, or shifts answer format.

2.4 Computational cost analysis

The paper states that the approach introduces additional computational cost due to multiple generations and paraphrasing and notes that steps can be parallelized to ensure $O(1)$ latency complexity. However, no empirical cost breakdown is reported, such as number of API calls, latency distributions, or monetary cost per query under typical k values and paraphrasing settings. This limits assessment of deployment feasibility under strict latency or budget constraints.

3 Technical Bottlenecks

3.1 Bottleneck 1: Reliably isolating epistemic uncertainty via perturbations

The central technical premise is that epistemic uncertainty manifests as instability under minor perturbations. Operationalizing "minor" without semantic drift is difficult for LLM prompts. Paraphrasing can change pragmatic presuppositions, entity references, or the question intent, while dummy tokens and system messages can alter style or policy behavior. Without verified semantic invariance, instability conflates epistemic uncertainty with input distribution shift.

3.2 Bottleneck 2: Output comparison for free-form generation

Inter-sample aggregation depends on a similarity function $s(\cdot, \cdot)$. For open-ended tasks, similarity metrics exhibit well-known trade-offs:

- lexical metrics (such as RougeL) reward surface overlap and can penalize valid paraphrases,
- embedding-based metrics reward semantic proximity but can blur factual inconsistencies,
- task-dependent answer formatting differences can distort similarity independent of correctness.

The study observes that RougeL often outperforms other tested metrics for calibration, which suggests that aggregation behavior is sensitive to metric bias. This metric dependence is a structural bottleneck because calibration quality becomes coupled to the idiosyncrasies of a similarity metric rather than correctness.

3.3 Bottleneck 3: Hyperparameter sensitivity and compute trade-offs

Calibration improves with the number of perturbed samples and plateaus around $k \approx 5$, indicating diminishing returns. Each additional sample increases inference cost. Temperature perturbation shows directional effects where increasing temperature improves calibration but random temperature perturbation does not. This indicates that effective calibration relies on carefully structured perturbations rather than generic stochasticity, creating a trade-off between robustness, tuning complexity, and compute.

3.4 Bottleneck 4: Dependence on auxiliary models and coupled uncertainty

Paraphrasing is produced via an LLM call. This introduces an additional stochastic component and potential systematic biases in the paraphraser. The uncertainty estimate becomes coupled to the paraphraser behavior, not solely the target LLM. Without controlling paraphraser quality, perturbation quality becomes a limiting factor for epistemic signal extraction.

4 Research Implications

4.1 Implications for LLM reliability and deployment

The findings reinforce that overconfidence persists even when outputs appear stable under naive resampling. Instability under meaning-preserving perturbations is an actionable signal for deployment, indicating that robust uncertainty estimation for LLMs requires probing sensitivity, not only sampling variability. The work also highlights an operational constraint: practical UQ methods must function under closed APIs, motivating input-level strategies.

4.2 Implications for evaluation practice

ECE-based calibration evaluation in LLM settings depends critically on how correctness is defined, especially for generation tasks. The study demonstrates that adapting sampling-based UQ to text generation necessitates output comparison protocols beyond exact match, elevating similarity metric choice from a minor implementation detail to a central methodological decision. This suggests that benchmark design for UQ should standardize both correctness definitions and similarity-based aggregation choices, or else calibration comparisons become protocol-dependent.

4.3 Connections to broader robustness and uncertainty literature

The approach connects LLM uncertainty estimation to adversarial robustness concepts by treating perturbation sensitivity as a proxy for epistemic uncertainty. This bridges two lines of research that are often separated: calibration and robustness. The observed benefits of structured perturbations indicate that LLM UQ can benefit from the long-standing methodology of controlled perturbation analysis, provided semantic validity constraints are enforced.

5 Potential Research Directions

5.1 Direction 1: Semantics-preserving perturbation protocols with verification

A concrete direction is to formalize perturbation validity with automated checks:

- entailment-based filtering using Natural Language Inference to ensure paraphrases preserve the original meaning,

- entity and relation consistency checks to avoid altering factual referents,
- template-based perturbations that vary syntax while preserving logical form for question answering.

Such protocols would reduce semantic drift, sharpening the interpretation of instability as epistemic uncertainty.

5.2 Direction 2: Task-aware aggregation beyond generic similarity

Aggregation can be made task-conditional:

- canonicalization for structured outputs (yes/no, multiple choice, numeric answers) before similarity,
- decomposition-based aggregation that compares extracted answers and rationales separately,
- learned calibrators that map agreement patterns and perturbation metadata to confidence, trained on held-out calibration sets.

This would reduce dependence on any single similarity metric and align aggregation with task semantics.

5.3 Direction 3: Cost-aware UQ with adaptive sampling

A practical direction is to reduce cost while preserving calibration:

- sequential decision rules that stop sampling once confidence stabilizes,
- bandit-style allocation across perturbation types, prioritizing perturbations that maximally increase disagreement when epistemic uncertainty is present,
- caching paraphrases and reusing perturbation sets across similar queries.

This directly addresses the absence of measured cost trade-offs by designing algorithms whose objective explicitly includes inference budget.

5.4 Direction 4: Extending evaluation to open-ended deployment settings

The limitations section identifies conversation and content generation as missing settings. Extending evaluation requires:

- operational correctness proxies for open-ended outputs (factuality checks, groundedness, or human preference labels),
- selective prediction evaluation (risk-coverage curves) to quantify how confidence supports abstention,
- stress tests with complex system instructions to quantify sensitivity to realistic prompt scaffolding.

5.5 Direction 5: Decoupling paraphraser influence

Paraphrasing introduces external model dependence. A direction is to characterize and control paraphraser-induced uncertainty by:

- using multiple paraphrasers and measuring variance attributable to paraphraser choice,
- generating paraphrases with constrained decoding and diversity controls to balance coverage and semantic fidelity,
- learning perturbation distributions offline on curated prompt pairs to reduce runtime paraphraser calls.

6 Conclusion

The paper presents a practical uncertainty quantification framework for LLMs that unifies sampling-based aleatoric signals with perturbation-based epistemic probes and introduces aggregation mechanisms that extend beyond exact match. Methodological strengths include an API-feasible perturbation design, systematic exploration of perturbation and aggregation options, and calibration-focused evaluation across multiple models and datasets with robustness checks for hyperparameter tuning. The most critical limitations are restricted evaluation to tasks with easily assessed accuracy, under-specified correctness definitions for summarization calibration, indirect ablation evidence, limited failure analysis, and the absence of empirical cost reporting despite increased inference requirements. The dominant technical bottlenecks arise from guaranteeing meaning-preserving perturbations, defining reliable agreement measures for free-form generation, and managing hyperparameter and compute trade-offs. The most promising research directions emphasize verified semantics-preserving perturbation protocols, task-aware aggregation strategies, adaptive cost-aware sampling, extension to open-ended deployment settings, and explicit decoupling of paraphraser influence from the target LLM uncertainty estimate.